

Spatial Intelligence for Urban Greening: Identifying Priority Locations and Greening Interventions

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Problem Description

- Nation wide greenery integration effort includes SG Green Plan's City in Nature.
- Ongoing campus-wide tree planting efforts on NUS campus since 2020.

Problem:

Finding suitable green and non-green areas for the integration of both trees and VGS on the campus



Related Work

Chen et al. (2026) studied the relationship between urban morphology, greenery, and outdoor thermal conditions in Singapore, which is closely related to the focus of our project.

Machine learning-based design optimization of urban greening for cooling in tropical cities

Chen et al. (2026) – Singapore city scale

Urban morphology + greenery



GPBoost surrogate model



UTCI prediction + SHAP interpretation



Greening optimization

Insight:

1. Built morphology and greenery both contributed to thermal prediction.
2. Tree fraction was one of the important predictors.
3. Greenery responses interacted with building morphology.

Research Gap

Machine learning-based design optimization of urban greening for cooling in tropical cities

Chen et al. (2026)	Our Project
Singapore city scale	NUS campus scale
200 m × 200 m grids	Monitoring-station surroundings
Physics-model-derived UTCI	Observed station air temperature
Morphology + greenery → UTCI	Weather + built → temperature vs. weather + built + greenery → temperature
Citywide optimization	Campus proof of concept

Data

The following dataset and tool will be considered for this study,

Data

- NUS BEAM:
for extraction of climatic variables
- STEVE Tool :
for extraction of location specific climatic & greenery variables

Data coverage – NUS Campus Meteorological Network

- Data from 40 weather stations spread across NUS were obtained with the following variables:
 1. Air Temperature
 2. Relative Humidity
 3. Atmospheric Pressure
 4. Global Irradiance
 5. WindSpeed
 6. Wind Direction
 7. Total Rainfall

Data Quality

1. Statistical test to identify potential outliers without any physical assumption
 - Repeated identical values which flags out as a frozen sensor
 - If more than 50% of night readings exceed 50 W/m^2 , they are removed for the month as they suggest sensors may be faulty
2. Using subject matter knowledge to decide which flag readings to remove physically impossible readings

Variable	Limit	Reason
Solar Radiation at night	$>50 \text{ W/m}^2$ at night is filtered	There should be no sunlight at night
Solar Radiation during the day	$>$ BSRN formula limit during the day	Using BSRN formula to limit what is more than physically possible for solar radiation
Pressure	Should not stray too far from 1000hPa	Singapore is on the equator

Data Quality

Solar sensors: 5 stations were flagged out to have solar faults.

- WS32,WS39 are dead sensors due to the lack of response to change in weather conditions.
- WS23 have captured abnormally high readings of around 1,500 W/m².
- WS07 and WS26 have unstable offsets from October to December where night readings were greater than 0W/m².

Pressure Fault: WS17 had faulty pressure readings from January to May.

Incomplete year round readings: Stations WS38,WS06,WS17,WS31,WS20 and WS30 were flagged out to having missing hours of data.

	valid readings	flagged	% flagged	most-flagged stations
variable				
temp	335381	9946	3.0	WS20 (36%), WS22 (11%), WS27 (11%)
rh	335381	5058	1.5	WS20 (23%), WS22 (11%), WS03 (3%)
wind_speed	335381	22267	6.6	WS16 (64%), WS02 (63%), WS35 (44%)
pressure	335381	2868	0.9	WS17 (33%), WS01 (0%), WS22 (0%)
solar	325335	27135	8.3	WS39 (66%), WS32 (59%), WS26 (45%)

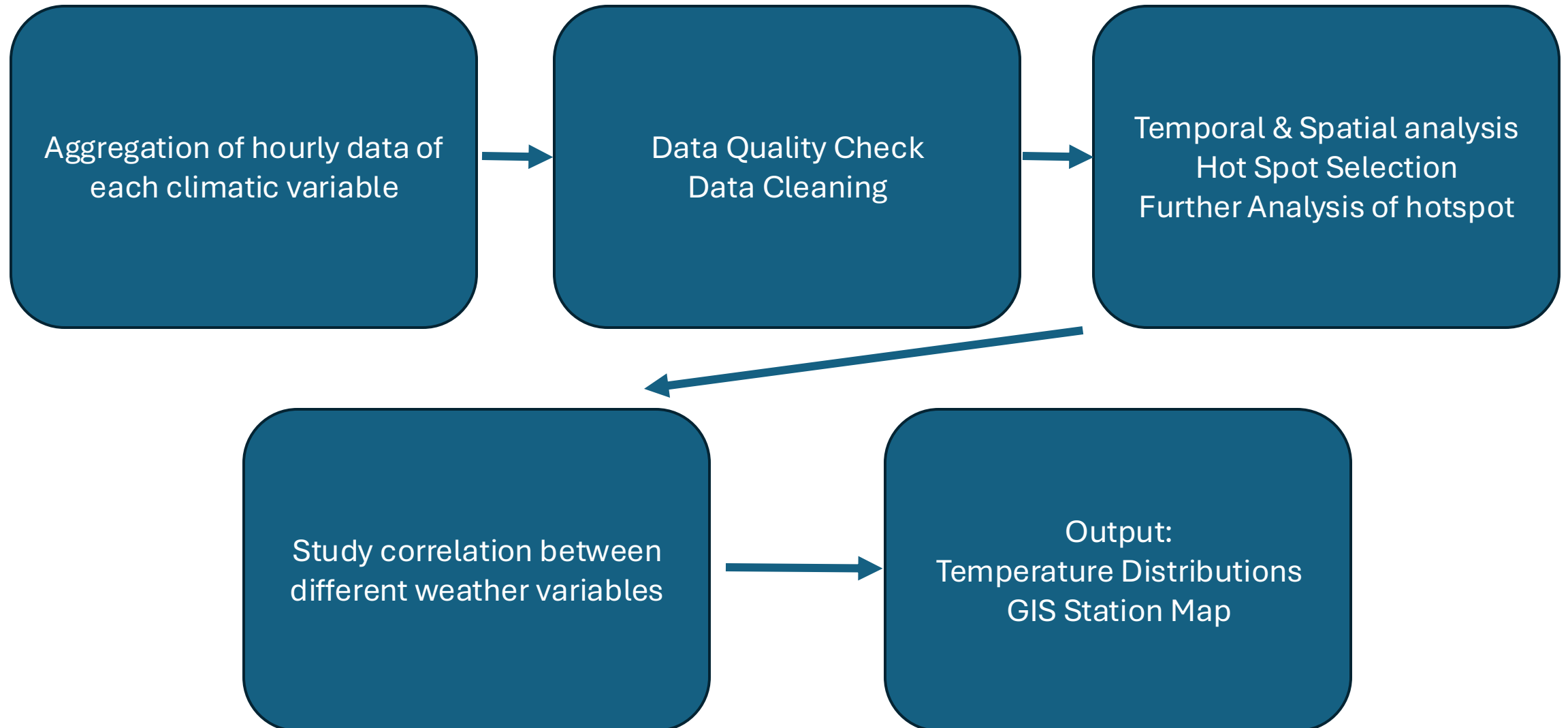
	readings in frozen runs	stations
variable		
temp	0	
rh	18	WS24 (18)
wind_speed	143	WS17 (143)
wind_dir	147	WS17 (147)
pressure	0	
solar	75	WS32 (50), WS07 (19), WS08 (6)

EDA

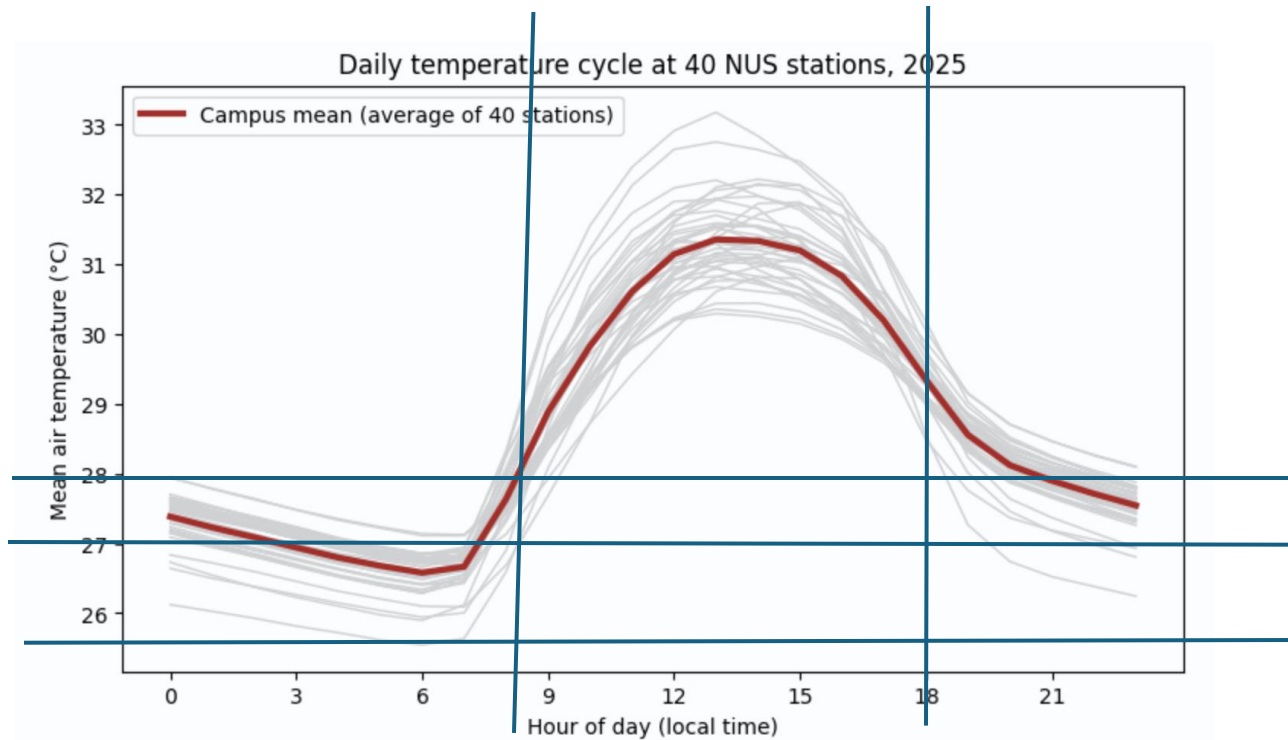
The objectives of EDA was to assess:

- What data do we have
- Is it usable for our purposes of our study and
- understand how do campus thermal conditions vary across space and time

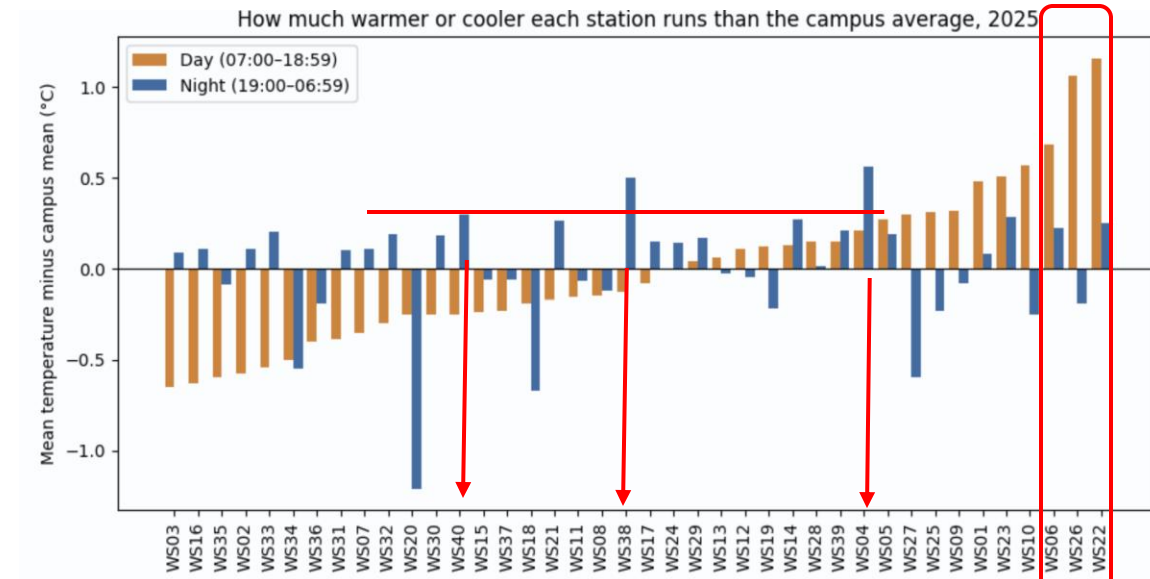
EDA Workflow



Key Insights from EDA Analysis

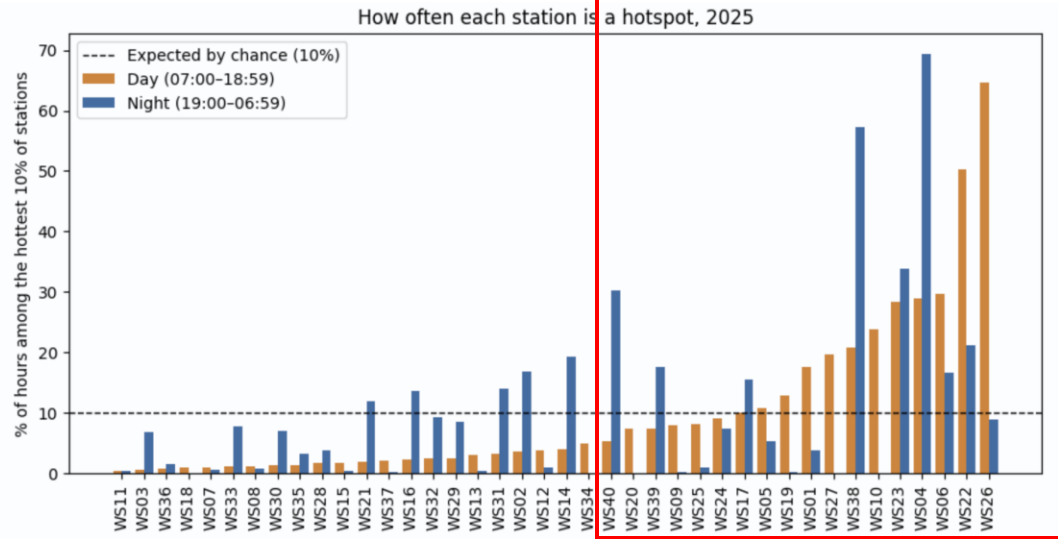


- During office hours, between (7am to 6pm), campus mean of the 40 WS ranges between 28 to 29.3°C
- After office hours, it ranges between 26.5 °C to 28°C

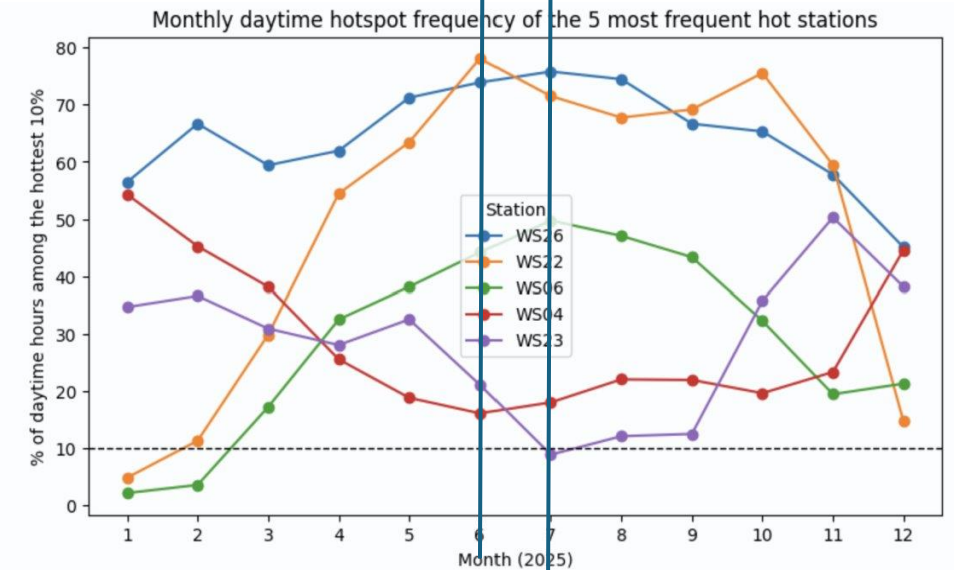


- Top 3 daytime hotspots are WS06, WS26 and WS22.
- Top 3 night-time hotspots are WS04, WS40 and WS38.

Key Insights from EDA Analysis



- Most frequent daytime hotspots in 2025 would be WS06, WS22 and WS26
- Most frequent nighttime hotspots would be WS04, WS38 and WS40.



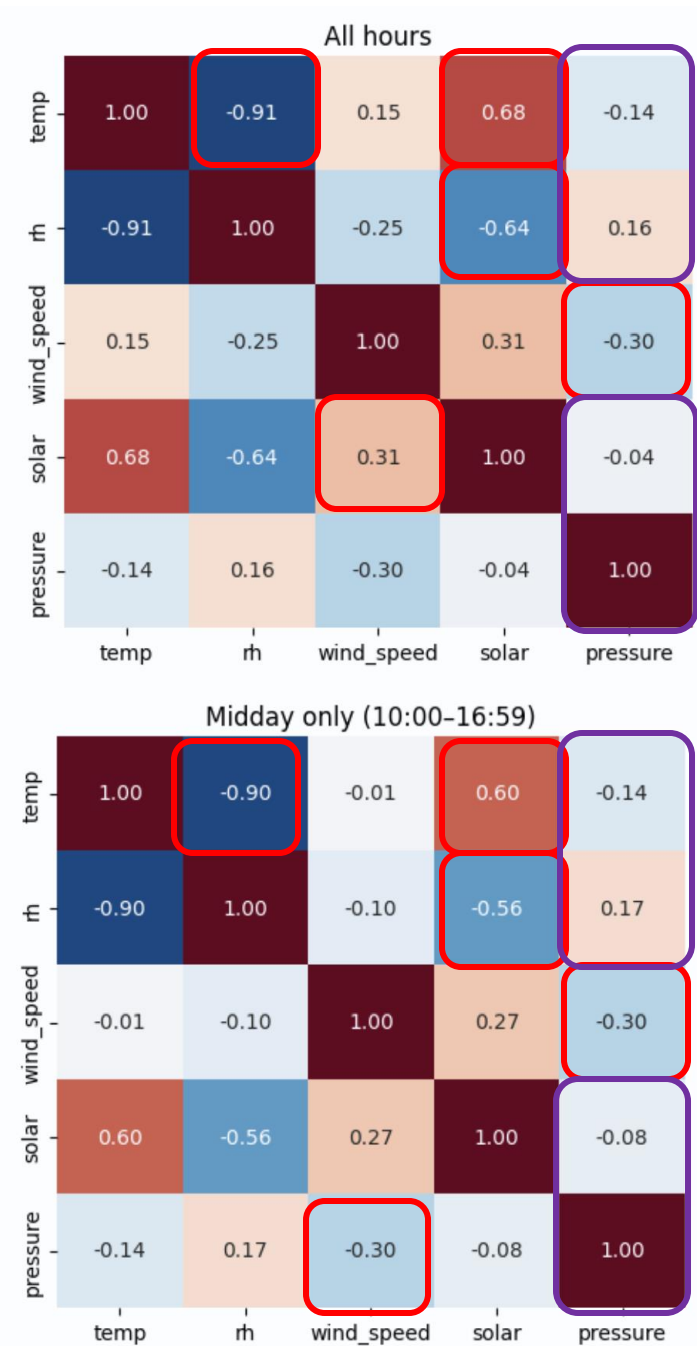
Of the 5 most frequent hotspots, WS26, WS22, WS06, there is a trend of is flagged out as hotspots more frequently between June-Nov, which corresponds to SW Monsoon & Inter-Monsoon 2 period.

Inversely, WS04 and WS23 hotspot's frequency occurrence is higher from Jan to May, which corresponds to NE Monsoon & Inter-Monsoon 1 period.

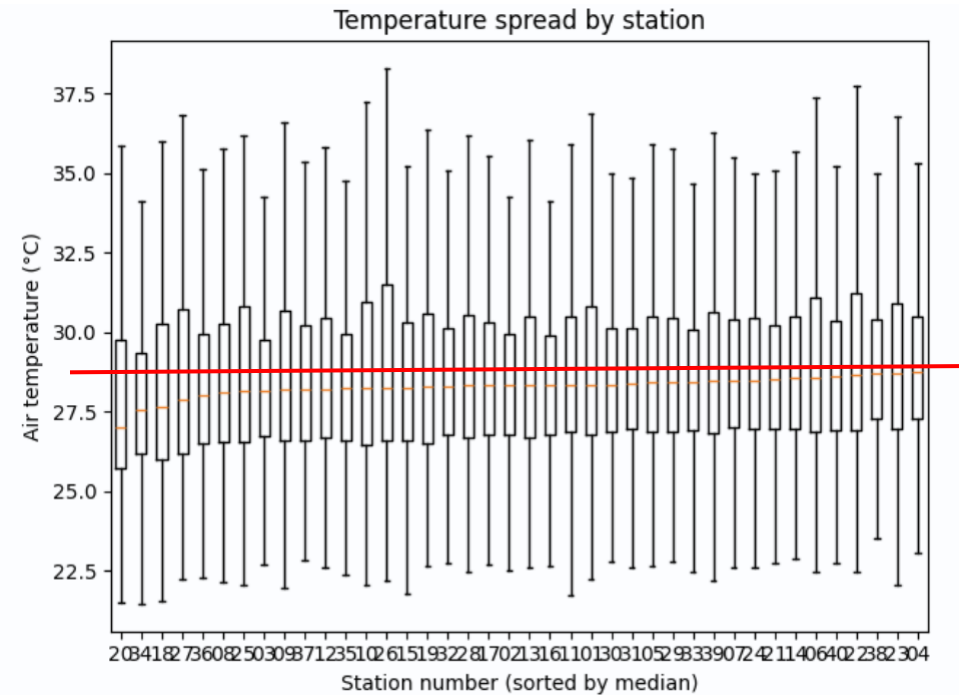
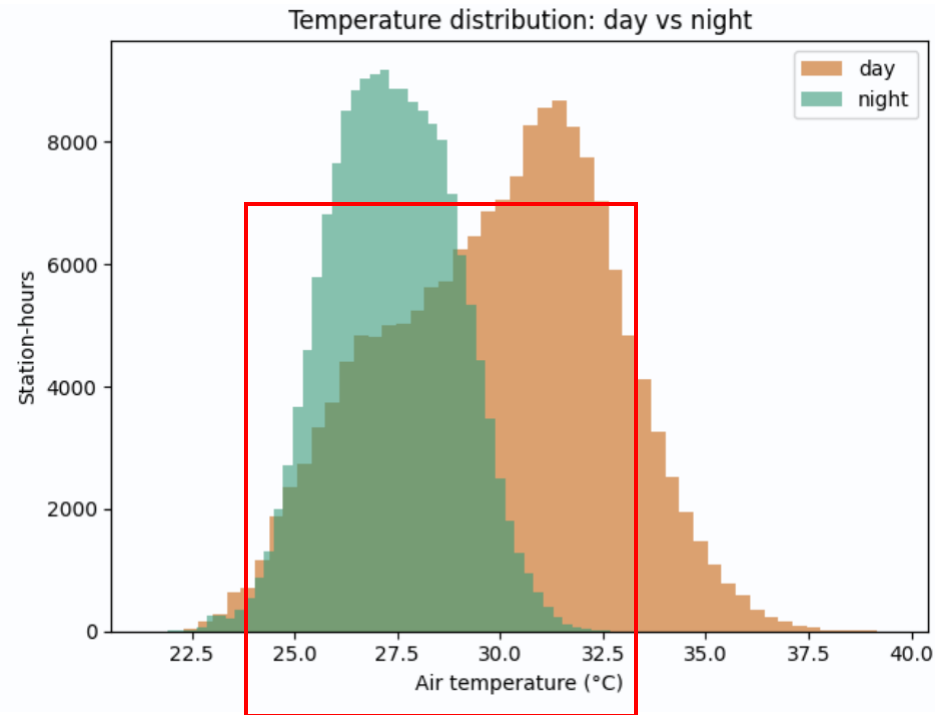
Key Insights from EDA Analysis

Variables	R values	Correlation	Conclusion
Temp & RH	$r = -0.91$ & -0.9	Strongly negative	Realistic co-relation. The two share an inverse relationship. Suggests drier and warmer seasons observed during periods of exposure to high Solar Radiation(Solar R.).
Solar R. & Temp	$r = 0.68$ & 0.6	Strongly positive	
Solar R. & RH	$r = 0.64$ & -0.56	Strongly negative	
Wind & other variables	$r = 0.31$ (with SR) $r = -0.3$ (with pressure)	Weak positive, Weak negative	Weak relationships with SR and pressure, even weaker with RH and temperature
Pressure & other variables	$ r < 0.2$ (temp, SR, hum) $r = -0.3$	Weakest correlation with other variables	Weak relationship between pressure and other variables, strongest correlation wind.

Key takeaway: Subsequent analysis to focus on the following climatic variables: Temperature, Solar Radiation and Relative Humidity.



Key Insights from EDA Analysis

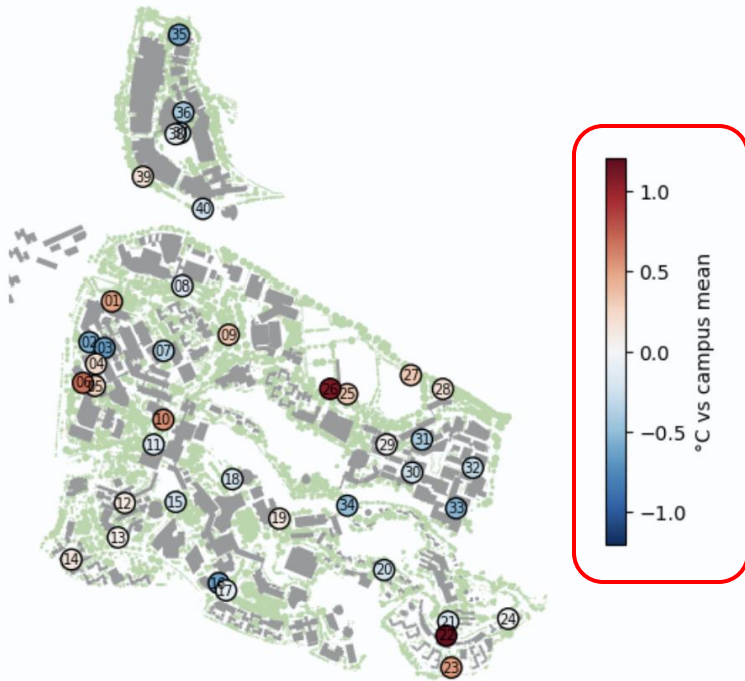


1. Overlaps in day vs night temperature distribution observed between 26 °C to 30 °C
2. Most stations have a similar temperature spread of air temperature by station
3. Gradual increase in median temperature observed over time, indicating some stations are warmer than other station, with peak median temperature being approximately 28.75°C
4. Some stations have wider temperature ranges and higher maximum temperature (36-38 °C), which suggest possible differences in local microclimatic conditions.

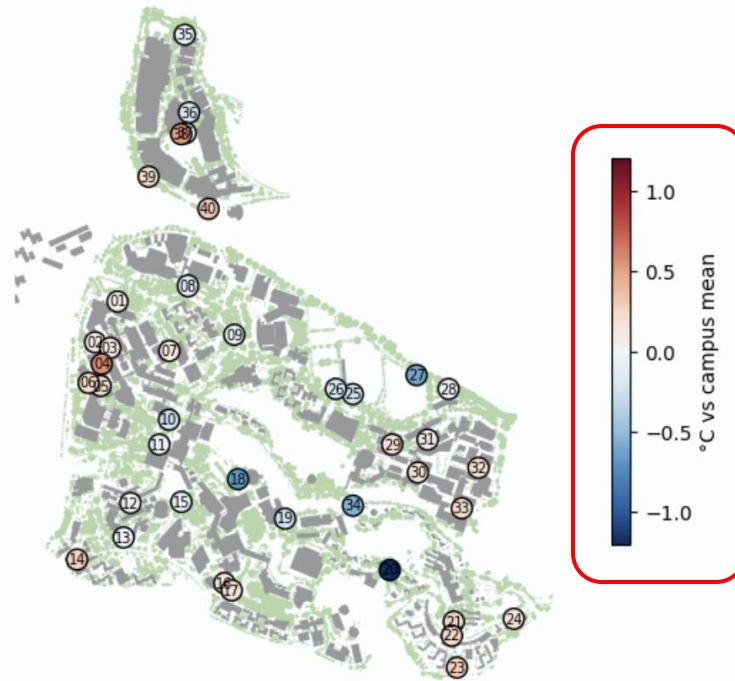
Key Insights from EDA Analysis

Station temperature anomaly over campus canopy (green) and buildings (grey)

Daytime (07:00-18:59)



Night-time (19:00-06:59)



1. Temp anomaly between canopy and buildings ranges between -1 to $+1$.
2. Daytime hotspot is WS26, while WS2-3, WS34, WS35 and WS37 show cooler than the campus mean, suggesting localized microclimate daytime conditions
3. Daytime hotspots may not be night-time hotspots. WS26 is one such example
4. Night-time hotspots include areas around WS33, WS39, WS40, WS21-WS23. These stations could be emitting stored heat from the daytime
5. Mix of hotspots and cooler areas observed in areas with existing greenery, hence suggesting other variables also influence the temperature anomalies.

Key takeaway: Hotspot or "Cold-Spots" are site-specific largely due to different microclimatic conditions.

Potential Method

1. Further EDA analysis to be conducted on required on greenery variables
2. A separate greenery variable would need to be subsequently characterized
3. Random Forest and XGBoost will be used to relate the surroundings of each weather station (tree canopy, building coverage, elevation) to its temperature anomaly relative to the campus mean, with solar radiation, humidity and wind as control variables. These 2 Tree-based models capture the non-linear relationships and interactions.
4. A linear regression will serve as a baseline.
5. Models will be tested on stations left out of training, and feature importance will then show which variables matter most.
6. Priority locations will be then consequently identified.

Sample Steps to use Tool

The following would be a sample of how the user could use the tool,

- User to place a tree on the model using a +Tree option
- The tool will ask the user to key in or select the greenery parameters.
Example of these parameters are as follows :
 - Tree height: 8m
 - Canopy: 6m
 - Leaf Area Index (LAI): 2.5
 - Tree Canopy Coverage: +10%

References

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